A-A-53210A
15 September 1989
SUPERSEDING
A-A-0053210 (DM)
30 September 1987

COMMERCIAL ITEM DESCRIPTION

RESIN, ACRYLIC, DENTAL, Inlay Pattern and RESIN, ACRYLIC, LIQUID, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a self-curing acrylic dental inlay pattern resin used in fabricating patterns for casting and performing soldering-related procedures.

Salient characteristics:

The items shall be of the following types.

Type I. - Combination package

Type II. - Standard package

Type III. - Replacement liquid

The resin shall be a red self-curing acrylic (autopolymerizing acrylate) dental inlay pattern resin suitable for use in fabricating patterns for the casting of inlays, crowns and bridges, posts and cores, and also suitable for use in performing soldering-related procedures. The resin shall be suitable for use in either the direct or indirect application with either the brush method or packed into or around the preparation in pre-mixed paste form.

Type I - Combination package. The combination package shall consist of a minimum of two ounces of red polymer powder, a minimum of two fluid ounces of monomer liquid, a minimum of 3.5 milliliters (ml) of lubricant, one dropper, three red sable brushes, two "Dappen" type dishes and instructions for use.

Type II - Standard package. The standard package shall consist of a minimum of two ounces of red polymer powder, a minimum of two fluid ounces of monomer liquid, a minimum of 3.5 ml of lubricant, a dropper and instructions for use.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:17Z
Check the source to verify that this is the current version before use.

Type III - Replacement liquid. The replacement liquid shall consist of a minimum of eight fluid ounces of clear, self-curing, acrylic resin liquid monomer.

Type II and Type III shall be compatible with Type I and each other.

Powder. The powder for Type I and Type II shall be a red polymer to contrast with the soft tissue and teeth in the mouth. The powder shall be very fine with a minimum of 70% of the powder passing through a #200 mesh screen.

Liquid. The liquid for Type I, Type II and Type III shall be a clear, self-curing acrylic resin monomer. Type III liquid shall be suitable as a replacement liquid for Type I and Type II.

Lubricant. The lubricant for Type I and Type II shall be petrolatum to facilitate removal of the resin pattern from the tooth cavity. The lubricant shall be red to contrast with the tooth structure and not discolor the resin pattern.

Dropper. The dropper of Type I and Type II shall be a three inch glass vial medicine dropper.

Brushes. The brushes of Type I shall have pure red sable tips. The tips shall be securely mounted in seamless metal ferrules and attached to a plastic or wooden, 5-1/2 inches minimum length, handle. The brushes shall be of the following sizes: one brush, size 00 and two brushes, size 3/0 (000). One of the size 3/0 brushes shall have its tip angled at 90^0 to enhance accessibility to the preparation.

Dish. The "dappen" dish of Type I shall be in accordance with the salient characteristics of Commercial Item Description A-A-51072.

Instructions. Instructions for Type I and Type II shall be complete and accurate.

Resin.

Physical characteristics. The resin shall set within six minutes, maximum. The dimensional stability shall be within $\pm$ .01%. The resin shall burn out completely resulting in a clean smooth mold, with no residue, in 40 minutes at 800^0 F (427^0 C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:17Z
Check the source to verify that this is the current version before use.

Performance characteristics. The resin shall produce a pattern superior to that of inlay wax. The resin shall closely adhere to the cavity walls of the tooth with less shrinkage, reproduce better detail and produce a more accurate impression than the corresponding inlay wax pattern. The resin shall have sufficient mechanical strength to allow for repeated trials. The set resin shall be sufficiently hard to prevent the loss of shape or distortion, after repeated drawings from the tooth cavity. The resin, when hardened, shall show no appreciable flaking or crumbling when trimmed or carved. The resin shall be compatible with inlay waxes such that these waxes may be added and will adhere to the set resin without adversely affecting the inlay wax or resin.

Color. Any color additive shall be approved and certified by the Food and Drug Administration under the Federal Food, Drug and Cosmetic Act.

Acceptability to ADA. The resin shall be listed as "acceptable" by the Council on Dental Materials, Instruments, and Equipment of the American Dental Association (ADA).

Irritation. Unless otherwise specified, prior to delivery the manufacturer shall furnish a statement directly to the Defense Personnel Support Center, DPSC-RSTS-2, Directorate of Medical Materiel, Technical Services Branch, 2800 South 20th Street, Philadelphia, PA 19101-8419 stating that the resin does not normally irritate oral tissues and does not contain poisonous ingredients in sufficient concentration to be harmful to human beings when used as directed.

Delivery schedule. Not more than three months shall have elapsed from the date of manufacture of the resin to the date of delivery to the Government.

Workmanship. The resin shall be free from lumps and foreign matter and any defects which may detract from its appearance or impair its serviceability.

Unit. Package (PG).

For Type I, one package containing resin acrylic, consisting of 2 oz (approximately 55 g) of red polymer powder, 2 oz (approximately 55 ml) of monomer liquid, 3.5 ml quantity of vaseline type lubricant, a dropper, three red sable brushes, 2 Dappen dishes and instructions for use, as specified, constitutes one unit. The powder, liquid, and lubricant shall be in suitable bottles. All components shall be packaged in a suitable box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:17Z
Check the source to verify that this is the current version before use.

For Type II, one package containing resin acrylic, consisting of two ounces of red polymer powder, two ounces of monomer liquid, 3.5 ml of lubricant, a dropper and instructions for use, as specified, constitutes one unit. The powder, liquid, and lubricant shall be in suitable bottles. All components shall be packaged in a suitable box.

Unit. Bottle (BT). For Type III, one bottle containing 8 oz (240 ml) of resin acrylic liquid, as specified, constitutes one unit. The bottle shall be packaged in a suitable box.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of resin or one bottle of replacement liquid. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, and sampling for tests (when applicable) shall be conducted at inspection level S-1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:17Z
Check the source to verify that this is the currentversion before use.

Examination. The dental acrylic resin shall be examined for defects including, but not limited to, those listed below.

Defects

Type I or Type II package does not contain all specified components.
Powder not free from lumps or foreign matter (Type I and Type II).
Lubricant not free from foreign matter (Type I and Type II).
Color of powder or lubricant not as specified (Type I and Type II).
Liquid not clear or has significant color.
Medicine dropper not proper size (Type I and Type II).
Sable brushes not as specified (Type I only).
Dappen dishes not as specified (Type I only).
Instructions for use missing (Type I and Type II).

Tests. Tests shall be conducted to determine compliance with specification requirements.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:17Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Type
6520-01-131-3438	ACRYLIC RESIN LIQUID, DENTAL, 8 ounces	III
6520-01-144-1517	RESIN, ACRYLIC, DENTAL, Inlay Pattern, Red	II
6520-01-151-1853	RESIN, ACRYLIC, DENTAL, Inlay Pattern, Red, Combination Kit	I

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:17Z
Check the source to verify that this is the currentversion before use.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2289
LOCATION: NCR DISK 2

58831
60094

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:17Z
Check the source to verify that this is the current version before use.